

Experiment No: 01

Date : _____

Roll No: _____

Aim : Draw horizontal, vertical, 30 degree, 45 degree, 60 and 75 degrees lines, different types of lines.

Theory:

Lines:

A line is a one-dimensional figure, which has length but no width. A line is made of a set of points which is extended in opposite directions infinitely.

In geometry, a line is a straight one-dimensional figure that does not have a thickness, and it extends endlessly in both directions. Diagram A represents a line. It does not have any endpoint. The two arrows at each end signify that the line extends endlessly and is unending in both directions. The length of a line cannot be measured. In simple terms, when a set of collinear points are connected in a one-dimensional plane, it is called a **line**.

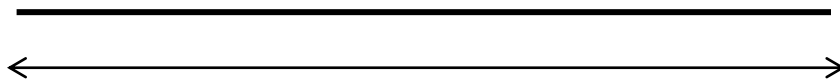


Fig: Line

line segment:

A line segment is a section of a line with specified ends. It has a start point and a finish point.

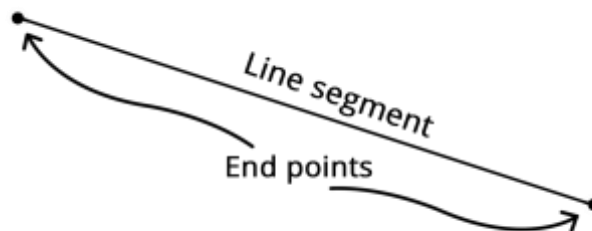


Fig: Line Segment

Different Types of Line :

1. Straight Line :

A line is a set of collinear points with no curves and extends limitlessly in opposite directions is called a Straight line.



Fig: Straight Line

2. Curved Line :

A line is a set of collinear points which has a curvature or bend in its formation is called a Curved Line.



Fig: Curved Line

3. Oblique Line

An oblique is a line that is diagonal or slanting. However, it is not perpendicular, parallel, vertical, or horizontal.

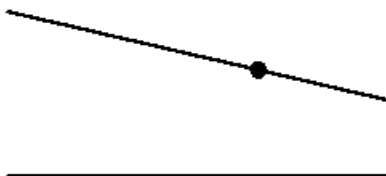


Fig: Oblique Line

4. Vertical Line:

A vertical line is a straight line that starts from the top to bottom or bottom to top. In simple terms, a vertical line is also known as a standing line. vertical is the opposite of horizontal. Anything that makes a 90-degree angle or a right angle with the horizontal line or the horizon is called vertical.



Fig: Vertical Line

5. Horizontal Line:

A horizontal line is a straight line that starts from either left to right or right to left. In simple terms, a horizontal line is also known as a sleeping line. A particular line at which the earth's surface and the sky appear to meet is called the horizon. Anything parallel to the horizon is known to be horizontal.



Fig: Horizontal Line

6. Perpendicular Line :

When two lines on the same plane intersect each other and form a 90° angle at the point of intersection, they are known to be perpendicular lines.

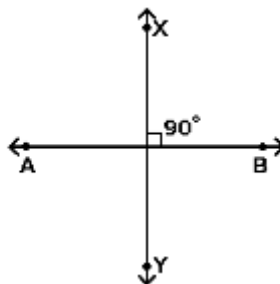


Fig: Perpendicular Line

7. Parallel Line:

A pair of two lines that are on the same plane and the distance between them is equal and remains constant. When such two lines do not intersect with each other, they are known as parallel lines.

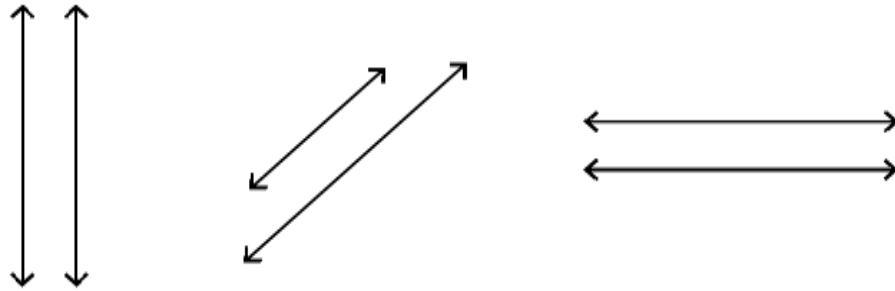


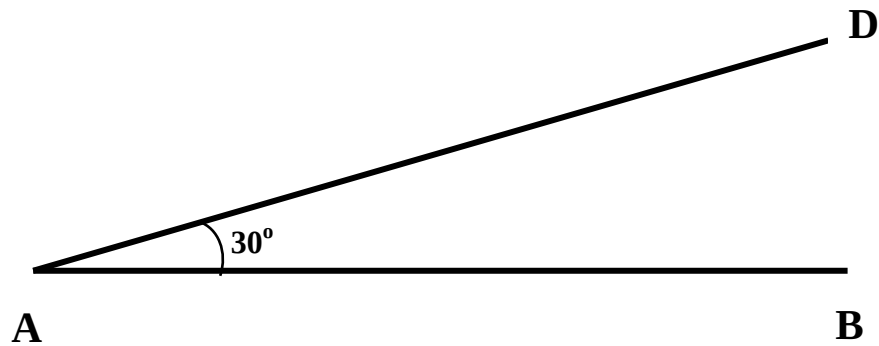
Fig: Parallel Line

Problem Statements :

1. Draw 30 Degree line
2. Draw 45 Degree line
3. Draw 60 Degree line
4. Draw 75 Degree line

1. Draw 30 Degree line

- Step 1: Draw a line segment AB.
- Step 2: Place the center tip of the protractor at point A such that the protractor perfectly aligns with line AB.
- Step 3: Start from 0 on the protractor in the anti-clockwise direction and stop at 30. ...
- Step 4: Plot point D and join with A.



Conclusion : Hence we drawn different types of lines.